Vol 2 Chapter 3 — Three Fermion Generations from Q^5 Cohomology

Keeper (author pass)

2026-05-23 Saturday

Chapter 3 — Three Fermion Generations from Q^5 Cohomology

The Standard Model has three families of fermions — three charged leptons (electron, muon, tau, with their neutrinos), three families of quarks (up/down, charm/strange, top/bottom). The number three is not derived in standard physics; it is empirical input. The masses of the fermions within each family span ten orders of magnitude (from ~ 1 eV neutrinos to ~ 170 GeV top quark) and are similarly empirical.

BST derives both: the *number* three and the *masses* of the second and third generations relative to the first.

3.1 Three generations from Q^5 cohomology

Lyra T1929 (the odd-power hypothesis) plus T1925 (rank forcing) together pin the number of fermion generations at three. The substrate-mechanism reading: the Hilbert polynomial of the quadric Q^5 at the relevant truncation gives exactly three K-type generations and no more. Increasing the truncation predicts additional generations not observed; decreasing it predicts fewer.

The three-generation structure is structural on the substrate. The substrate cannot accommodate a fourth generation without changing Q^5's cohomology — which is, of course, forced by the substrate's complex dimension $n_C = 5$ (the quadric in 5 complex dimensions is what we have).

3.2 The muon-to-electron mass ratio

The famous lepton mass ratio:

$$\frac{m_\mu}{m_e} = \left(\frac{24}{\pi^2}\right)^6 \approx 206.7682.$$

Measured: $m_\mu/m_e = 206.7682830 \pm 0.0000046$. BST prediction matches at 0.004% (Cal #100 cross-volume tier reconciliation corrected from earlier reported 0.05–0.06%, with the 0.004% canonical Friday May 22, 2026).

T190 anchors this in the substrate: $(24/\pi^2)^6$ uses the integer 24 ($= 4 \cdot C_2$, see Vol 4 Ch 1's Newton's-G derivation for the same integer's appearance) raised to the sixth power, divided by π^{12}. The factor of 6 in the exponent traces to the same $C_2 = 6$ structure that gives the proton-to-electron mass ratio $6\pi^5$ (Chapter 6).

3.3 The tau-to-electron mass ratio

$$\frac{m_\tau}{m_e} = 49 \cdot 71 = 3479,$$

with measured $m_\tau/m_e = 3477.15$. Match at 0.05%. The factorization $49 \cdot 71$ is structurally significant: $49 = g^2$ (gauge-dimension squared) and 71 is one of the Monster supersingular primes (Volume 4 Chapter 4's t_{cosmo} neighborhood). Lyra T2003.

3.4 Mass-hierarchy structure

The mass hierarchy across generations follows the Wallach K-type weight structure on D_{IV}^5. Higher generations correspond to higher K-type weights, with the mass ratios fixed by Casimir-eigenvalue scaling.

The full inter-generation mass spectrum derivation is in active development (multi-month K52a Sessions 10+ research). What is established at chapter-grade is: three generations forced; first-to-second mass ratio (m_μ/m_e) at 0.004%; first-to-third mass ratio (m_τ/m_e) at 0.05%; full hierarchy structure substrate-natural via Q^5 Chern-class spectrum.

3.5 What comes next

Chapter 4 — Color and Quarks — develops the $SU(3)$ confinement mechanism on the substrate as a Frobenius closure phenomenon.

Where to look this up: T190 (m_μ/m_e, Cal #100 corrected 0.004%). T2003 ($m_\tau/m_e = 49 \cdot 71$). T1929 (odd-power hypothesis for three generations). T1925 (rank = 2 forcing). For experimental status: CODATA 2018 lepton mass evaluations.